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Interpreting DNA Evidence: A Review

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Summary

The paper provides a review of current issues relating to the use of DNA profiling in forensic science. A short historical section gives the main statistical milestones that occurred during a rapid development of DNA technology and operational uses. Greater detail is then provided for interpretation issues involving STR DNA profiles, including:

- **methods that take account of population substructure in DNA calculations;**
- **parallel work carried out by the US National Research Council;**
- **the move away from multiple independence testing in favour of experiments that demonstrate the robustness of casework procedures;**
- **the questionable practice of source attribution 'with reasonable scientific certainty';**
- **the effect on the interpretation of profiles obtained under increasingly sensitive techniques, the LCN technique in particular;**
- **the use of DNA profiles as an intelligence tool;**
- **the interpretation of DNA mixtures.**

Experience of presenting DNA evidence within UK courts is also discussed. The paper then summarises a generic interpretation framework based on the concept of likelihood ratio within a hierarchy of propositions. Finally the use of Bayesian networks to interpret DNA evidence is reviewed.

Key words: Forensic science; Interpretation of evidence; Likelihood ratio; DNA profile; DNA mixtures; Bayesian networks.

1 Introduction

Over a period of 16 or so years, DNA profiling has grown, through a series of imaginative ideas and brilliant technological implementation, to become the most powerful scientific technique available to the crime investigator since the adoption of fingerprints as a means of personal identification at the turn of the 20th century. When a serious crime hits the national headlines, DNA profiling evidence is often expected to play a vital role in bringing the perpetrator to justice.

In many ways, DNA profiling has become a model for how new techniques should be introduced into the forensic arena. Many of the aspects of validation are technical but in this paper we address the core inferential issues. Some of these are statistical; some call for the study of population genetics models. For us, the most interesting relate to the fundamental questions of how scientific evidence should be set within the framework of legal proceedings so as to provide balanced guidance to a court or jury.

2 A Brief History of the Evolution of Statistical Issues Relating to DNA Profiling

Following the initial work by Jeffreys *et al.* (1985a, b), who pointed out the potential of the technique in forensic science, DNA profiling was first used in the UK in 1986 to provide powerful evidence in the case of *R v Colin Pitchfork*. The profiling method at that time was known as the Multi Locus Probe (MLP) technique. Results were visualised as a set of parallel bands on a photographic plate. The rather misleading term of DNA fingerprinting was coined for this method, e.g. Jeffreys (1985b).

MLPs were compared visually, rather than relying on any agreed statistical methods. Matching patterns were compelling and if sufficient numbers of bands were present, most practitioners reported that the evidence conclusively identified the samples as having come from the same individual. Problems soon arose with cases where only some of the MLP bands matched, cases involving relatives and application of the method to mixed samples or paternity work. There was also the issue of whether or not individual MLP bands were present in the profile. These 'biological' arguments were used in *R v Andrew Deen* in 1990, and the statistical interpretation was challenged in *R v John Henry Bell* in 1993 (both cases were tried in England). Another disadvantage of the MLP technique was that relatively large biological samples were required to give a reliable result.

The Single Locus Probe (SLP) technique was introduced to casework in the UK from about 1988. In the USA this has always been known as the Restriction Fragment Length Polymorphism (RFLP) technique. The probes were used to target specific regions of DNA. By using locations (known as loci) that were on different chromosomes, it was hoped that results obtained from the different probes could be treated as essentially independent. Each probe produced one or two bands, referred to as a homozygote and heterozygote respectively, depending on whether the versions of the DNA sequence an individual inherited from its two parents was the same or different at the given locus. The most widely used method of interpretation involved a two-stage approach. First, an assessment was made as to whether two profiles were close enough in size to be considered to match; this was then followed by an assessment of the significance of the match. A number of different methods were adopted for the two stages, including the use of fixed or sliding windows at the comparison stage, and the use of variable or fixed bins to assess the significance of the match. These methods are described in Gill & Werrett (1990) and Gill *et al.* (1990). Budowle *et al.* (1991) describes the SLP methodology adopted in North America. This was used as a model for many other DNA testing laboratories around the world

In the second stage of the interpretation, a database was required to give an estimate of the relative frequency in the population of each of the matching SLP bands. These were typically of the order 5–10% and a typical database size was about 200. In the UK, separate databases were constructed for the three main ethnic groups, representing people of Caucasian, Afro-Caribbean and Asian (Indo-Pakistani) descent. Making the assumption that the population was in approximate Hardy–Weinberg Equilibrium (HWE), which implies that bands are inherited independently within loci, and using about 4 different SLP probes, the technique was capable of producing very powerful evidence with match probabilities of the order of 1 in many millions. The match probability is an estimate that unrelated members of the population would have the same SLP profile. These were calculated using a method termed the product rule that essentially multiplied the relative frequencies of matching bands within and across loci.

The statistical tests used to establish the robustness of the approach are described in Devlin & Roeder (1990), Evett & Pinchin (1991) and Evett & Gill (1991). However, the probability model adopted, based on the product rule, was widely challenged; e.g. Lander (1989), Thompson & Ford (1990), Cohen (1990). Critical issues that were raised included: the size and representativeness of the databases; default band frequencies; independence of results at the different loci; measurement error and laboratory error; dealing with relatives; and mixed profiles from DNA that had come from more than one individual. The debate became extremely acrimonious in North America with opposing

arguments on the effects of population sub-structure being put forward by Chakraborty & Kidd (1991) and Lewontin & Hartl (1991). A good unbiased review of the debate is provided by Weir (1992).

A more unbiased continuous approach to evaluating SLP evidence was provided by Berry *et al.* (1992) and Evett *et al.* (1993). This avoided the need for the two-stage comparison; instead, the evidence was reported as a likelihood ratio (LR) rather than a match probability. The adequacy of the match probability and LR values resulting from the above methods was demonstrated by large-scale experiments carried out by Risch & Devlin (1992) and Lambert *et al.* (1995), involving millions of between-person comparisons.

Nichols & Balding (1991) highlighted the need to take account of population substructure in DNA calculations. This could be achieved by consideration of a term, denoted by F_{st} or θ, a measure of the genetic differentiation between populations. Their ideas were further developed in Balding & Nichols (1994) and expressions derived for the conditional probability that two individuals belonging to the same racial subpopulation would match:

$$\frac{2\,(\theta + (1-\theta)p_A)\,(\theta + (1-\theta)p_B)}{(1+\theta)(1+2\theta)}, \qquad \text{for heterozygote } (A, B)$$

$$\frac{(2\theta + (1-\theta)p_A)\,(3\theta + (1-\theta)p_A)}{(1+\theta)(1+2\theta)}, \qquad \text{for homozygote } (A, A)$$

where p_A and p_B denote the proportions of band A and B in the racial population of interest. These have been widely accepted in the forensic community, as has the sampling adjustment that they advocate to estimate the proportions of the matching SLP bands, p_A and p_B. This adjustment adds the crime and suspect profile temporarily to the database before estimation using database relative frequencies. This effectively provides an approximation to the Bayesian estimates when a trinomial or binomial likelihood is adopted along with uniform priors. The paper also provides information on the LR calculations relevant when relatives of the suspect are considered as alternative potential sources of the crime DNA, an issue first raised by Evett (1992).

The U.S. National Research Council (NRC, 1992) produced a report in 1992 that recommended employing a very cautious approach to the interpretation of DNA profiling evidence, which included using the "ceiling principle" to evaluate match probabilities. The document was widely criticized (e.g. Devlin *et al.*, 1993, 1994), prompting the production of a second report (NRC, 1996) which provided a much better statistical foundation on which to base interpretation of DNA profiles (discussions can be found in Weir, 1996; Balding, 1997; Koehler, 1997; Lempert, 1997; Morton, 1997; Thompson, 1997).

SLP evidence was frequently challenged in the courts (Redmayne, 1995). Kaye (1993) and Thompson (1993) provide reviews of litigation issues that arose in the US. One important issue that led to three successful appeals in the UK was the fallacy of the transposed conditional, known as the Prosecutor's Fallacy in the forensic context (Balding & Donnelly, 1994). This is the error of confusing the probability of the DNA evidence given a particular proposition with the probability of the proposition given the DNA evidence. In *R v Deen*, the Court of Appeal ordered a retrial because it realized that the prosecution's expert testimony may have led the jury to confuse a one in three million match probability with a one in three million probability that the accused had not committed the rape (*R v Deen*, *The Times*, 10 January 1994, CA No 90/1523/X3). The following example is from the transcript of the trial of Alan Doheny for rape and buggery in 1990.

FORENSIC SCIENTIST	*I calculated the chance of finding all these bands and the conventional blood groups to be about 1 in 40 million.*
JUDGE	*The likelihood of it being anyone other than Alan Doheny?*
FORENSIC SCIENTIST	*Is about 1 in 40 million.*

We note that, in contrast to the U.K., no U.S. court has reversed a conviction because the DNA match probability was misrepresented as the probability of innocence.

A slightly different error was made in the trial of Gary Adams for buggery in 1991. After the scientist had reported a match probability of 1 in 27 million, the trial judge in summing up stated

> *. . . I should think there are not more than 27 million males in the United Kingdom which means that it is unique.*

The Doheny and Adams Appeals were heard jointly and the subsequent Court judgment (*Doheny & Adams* [1997] 1 Cr. App. R. 369) included a number of recommendations for the reporting of DNA evidence, one of which was that it was not part of the scientist's role to comment on the likelihood that it was the defendant who left the crime stain. Although conclusive opinions are still routinely given in other areas of forensic evidence, it is now widely accepted among DNA practitioners that consideration of the prior and posterior odds in a case is the domain of the jury. Evett (1995) provides advice on how to avoid the transposed conditional with some examples.

DNA profiling systems were further refined through development of the Short Tandem Repeat (STR) technique in 1993. This followed on from the pioneering work of Mullis who invented a technique (called PCR) that facilitates the profiling of small quantities of DNA. STR regions usually consist of a specific core sequence of chemical components which is repeated about 5–30 times. There are many hundreds of these STR regions in the human genome, but 10–15 are sufficient to give exceedingly high levels of discrimination between individuals. Resulting STR profiles appear as a series of peaks on a graph; each peak is labeled with a specific number according to its position on the graph. This represents the number of times the particular core sequence is present. These numerical designations are known as *alleles*. The STR-PCR technique offered some clear advantages over the SLP system:

- The size of the STR peaks can be measured accurately and unambiguously. Therefore, STR profiles are represented simply as a series of numbers (alleles), two at each locus inherited from each parent. This has facilitated the construction of intelligence databases containing profiles from undetected crimes and known criminals.
- Several STR regions can be amplified at the same time; this was not possible with SLPs.
- Smaller DNA samples can be analysed.
- There is a large number of STR regions in the human genome so there is no theoretical limit to the discriminating power of the technique.

Gill (2002) recently published a review of the development of STRs for forensic use; we will concentrate here on the statistical aspects.

The quadruplex (quad) was the first STR system to be introduced at the Forensic Science Service (FSS). Quad contained four loci (Gill & Evett, 1995) and this was followed by the SGM system that analysed 6 loci (Evett *et al.*, 1997). The current SGMPlus system is based on 10 loci (Foreman & Evett, 2001) (plus a test that indicates the sex of the donor) and is used to provide profiles for entry on the National DNA Intelligence Database.

Introduction of the STR technique did not lead to any new statistical issues being raised in court. Concerns still remained as to what databases were used to estimate the allele proportions. Match probabilities obtained for the quad system were of the order 1 in 10,000 and standard significance tests, such as χ^2 goodness-of-fit and Fisher's exact test (Zaykin *et al.*, 1995), were used to justify independence assumptions in the calculations (Evett *et al.*, 1996).

Evett & Buckleton (1996) discussed the fallacies inherent in significance testing for independence. In essence, the process is fundamentally flawed because it invokes the idea of testing a null hypothesis of perfect independence (either within- or between-locus) that cannot be true in real populations. The issues are also discussed in Evett & Weir (1998) and in Buckleton *et al.* (2001). It was at this time

that the FSS moved away from the idea of multiple independence tests in favour of experiments that demonstrated the robustness of casework procedures. Furthermore, Balding & Donnelly (1995a,b), Balding (1995) show that *the product rule estimator has undesirable sampling properties* and provides justification of the sampling adjustment described earlier. This is achieved through Bayesian estimation of products of allele proportions using a binomial or trinomial likelihood and uniform priors. Foreman *et al.* (1997, 1999) provide a similar argument using a more general multinomial likelihood for the entire vector of allele proportions. A recent analysis of various sampling allowances (Curran *et al.*, 2002) showed that Balding & Nichols's proposed sampling adjustment provides a good approximation for reasonable-sized databases.

There were a number of UK court cases around this time in which match probabilities were calculated for more than one analysis technique; e.g. SLP and STR, STR and conventional blood grouping. The results were not combined numerically because of the limited quantity of data relating to the independence of the techniques. In 1993, the case of *R v Shatanawi* included evidence from several techniques which were treated separately, whereas assuming independence and multiplying the results together in *R v Doheny* provided one of the grounds for the subsequent successful appeal (*Doheny & Adams* [1997] 1 Cr. App. R. 369).

However, in the U.S. information from various systems *was* combined by assuming linkage equilibrium. The fact that more than one technique was used to type various loci did not lead to any more successful challenges than those with regard to combining the results of distinct loci typed using a single technique.

The trial of Dennis Adams for rape in 1995 was unusual in that the prosecution case depended almost entirely on the DNA evidence. Consequently, the defence proposed the use of Bayes' theorem to combine the DNA evidence with the non-DNA evidence, which supported the view that Adams was not guilty (an alibi put forward by Adams and the fact that the victim did not recognize him as her offender). Adams was convicted but the Court of Appeal ordered a retrial after stating that the judge had misdirected the jury in the use of Bayes' theorem (*Dennis Adams* (No.1), *The Times*, 3 November 1997 CA No. 96/6927/Z5). At the retrial, a questionnaire was produced to 'help' the jury use Bayes' theorem to combine the various pieces of evidence. Adams was re-convicted and at the second unsuccessful appeal, the Court stated that the questionnaire was seen as a recipe for confusion (*Dennis Adams* (No.2) [1998] Cr. App. R. 377). This poses something of a dilemma as the use of Bayes' theorem is seen by most of those who have made serious study of inference in the legal context as the most logically satisfactory way of combining evidence (Robertson & Vignaux, 1997, 1998).

A new Bayesian approach for examining STR databases was developed by Foreman *et al.* (1997, 1999). Estimates of F_{st} were obtained for the 3 major UK ethnic groups using FSS quad frequency databases and in the absence of any specific subpopulation data. This was treated as missing data and a MCMC (Markov chain Monte Carlo) approach adopted to sample from the complex posterior distribution involved. The availability of SGM data at subpopulation level enabled more detailed investigations of population substructure to be carried out, as described in Foreman *et al.* (1998) and Foreman & Lambert (2000). A Bayesian approach was again adopted using the recursive equation detailed in Balding *et al.* (1996) to obtain joint probabilities of alleles in a racial subpopulation. This equation holds when subpopulation allele proportions are assumed to follow a Dirichlet distribution with parameters dependent on the racial group allele proportions, p_A, p_B, ... and F_{st}.

Roeder (1994) provides a comprehensive review of the statistical issues involved with SLP profiling and discusses the major points of controversy. This is followed by a later review that also addresses STR methods (Devlin & Roeder, 1997). More recently, Evett & Weir (1998) describe the probabilistic reasoning necessary to evaluate simple single-stain criminal cases and the more complex calculations involved when considering DNA mixtures from more than one person and parentage analyses. These will be addressed in greater detail later in the paper.

3 Statistical Evaluation of SGMPlus and LCN Profiles

Since their introduction in the early 1990's, STR profiling techniques have continued to improve, primarily in terms of increased sensitivity so that profiles can now be obtained from minute amounts of starting DNA. The most recent technology is referred to as Low Copy Number (LCN) because of the sparsity of DNA cells available to the PCR process for amplification. We discuss this approach later in the section.

3.1 Profiles Obtained from the Standard SGMPlus Technique

As the STR profiling systems evolved, improvements were also seen in terms of increasing forensic discriminating power; each new system would consider a greater number of loci than the previous one. As discussed earlier, in the FSS, the quad system (4 loci) was followed by the SGM system (6 loci) and, more recently, the SGMPlus system (10 loci).

Match probability calculations under the quad and SGM systems were carried out using the approaches discussed earlier and advocated by Balding & Nichols (1994). These had become widely used among forensic organisations throughout the British Isles, Europe, Australia and New Zealand although in the U.S., the NRC recommends their use only when there is specific case information that potential sources of the DNA originate from some well-defined racial sub-group.

However, with the introduction of the SGMPlus system, it was clear that the standard calculation would give rise to very small match probabilities, certainly for full and "almost-complete" profiles.

In the U.S., the FBI uses the CODIS system comprising 13 STR loci. They consider that *objective criteria can be used to report that with reasonable scientific certainty a particular individual is the source of an evidentiary sample* (Smith & Budowle, 1998; Budowle *et al.*, 2000); i.e. a conclusive opinion as to the source of the DNA is given. Practitioners who follow a Bayesian approach to evidence interpretation, which include staff in most forensic organisations in the British Isles, Eire, much of western Europe and New Zealand, believe that this necessarily invokes Bayes' theorem and an assessment of the prior odds. This is generally accepted to be a decision for the jury, not the scientist. Foreman & Evett (2001) do not support this approach since the method of calculation of match probabilities cannot be checked by the necessary between-person comparisons described earlier for SGM profiles. They investigated the magnitude of match probability figures for full SGMPlus profiles and a range of categories of alternative suspect, from a full sibling of the suspect to an unrelated person. A general, non-case-specific approach is advocated based on figures arising from the theoretically most common SGMPlus profile. At the present time, it is the policy of the FSS in those cases where there is a match between two full SGMPlus profiles not to carry out a match probability calculation for unrelated people: instead the figure of one in a billion is given—whatever the ethnic group cited in the alternative proposition. Foreman & Evett (2001) argue that this is a fair and reasonable approach given the current state of knowledge of dependence effects. Calculations for partial profiles and mixtures are subject to the condition that a billion is the largest likelihood ratio that will be quoted.

Issues surrounding the reporting of SGMPlus evidence in court are discussed in Evett *et al.* (2000a). These include recommendations made by the Appeal Court in the cases of Alan Doheny & Gary Adams; i.e. to provide an estimate of the number of people who would match the crime profile in some specified population. A further approach considered relates to the idea of uniqueness of the crime profile. Balding (1999) derives an upper bound for the probability that the suspect profile is unique within some pre-specified population containing blood relatives of the suspect in addition to unrelated people.

Challenges to the statistical evidence have become rarer with SGMPlus profiles. The crucial issues questioned by defence now relate to: how & when the DNA was deposited; its relevance to the alleged crime (i.e. consideration of activity-level rather than source-level propositions—see Section 5); establishing continuity of exhibits & integrity of the evidence; and laboratory error.

It is worth mentioning here the importance of investigating how to present DNA evidence in court. It has been shown that scientists and lay people often experience difficulties with probabilistic arguments (Taroni & Aitken, 1998).

More recently, Koehler (2001a, b) discusses experiments using mock juries with the objective of identifying psychological factors that may influence a juror's understanding of the DNA evidence in a case. In straightforward cases involving a single crime stain matching the defendant, it was found that relative frequencies were better understood than probabilities although this effect became less pronounced as match probability values decreased. Furthermore, it was found that reporting minute match probabilities, of the order 1 in several billions or less, did not result in jurors assigning 100% probability to the defendant being the DNA source; perhaps this was due to their considering the possibility of laboratory error.

Hoffrage *et al.* (2000) suggest translating match probabilities into frequencies by considering the match probability in the context of an initial large population. Cooke made the same suggestion arguing that this mode of presentation prevents contravention of the Doheny and Adams court of Appeal ruling (Cooke, 2000, 2001). Although it may seem appealing to report DNA evidence in the form of absolute frequencies, the fact that match probabilities depend heavily on relatedness makes systematic use of frequencies less than helpful, if not misleading (Evett, 2001).

3.2 Issues Surrounding the LCN Technique

More recently, the LCN technique has been introduced into casework at the FSS (Gill, 2001). This is extremely sensitive and enables profiles to be obtained from only a few cells of starting DNA. In addition to considering issues of transfer and persistence of DNA under different circumstances (e.g. Lowe *et al.* (2002) and Wickenheiser (2002)), adequate interpretation of LCN profiles will involve modelling contamination and so-called drop-in of spurious alleles; also, determination of the reproducibility of profiles. Gill *et al.* (2000) attempt to provide a probabilistic interpretation of LCN profiles but there is currently a lack of relevant data to inform rigorous statistical modelling of these profiles.

Many of the factors that affect profiling are seen to a greater degree in LCN analysis and these exacerbate the limitations of the models available. These factors include sampling variation due to the low number of DNA cells available in the original sample, leading to incomplete representation of alleles from the biological sample in the final profile (often called 'allelic drop-out'). This makes the designation of a single allele at a locus as a homozygote less reliable. The LCN technique increases the intensity of the alleles that have been detected to well above baseline on the graph but imbalances observed between heterozygote alleles is also increased. Under the standard SGMPlus system, true heterozygote alleles originating from one individual have similar intensities; i.e. similar peak heights and areas. Other artefacts of the profiling technique are more pronounced with the LCN technique and during pre-processing it can be more difficult to assess whether or not a peak should be designated as part of the profile. Stochastic contamination can occur when a cell (or cells) is introduced into the reaction, showing as several additional alleles in the profile. Additionally, a small fragment of DNA containing a single allele may be introduced into the reaction, showing as a single additional allele in the profile. It is possible that this DNA fragment comes from degraded human epithelial cells (i.e. dust particles) or it may itself be the product of an earlier PCR amplification.

The practical implications of some of these issues are explored by Whitaker *et al.* (2001), who also describe how the LCN profile is based on a consensus derived from duplicate reactions. When all or most alleles can be reproduced then the consensus is taken to be a reasonable representation of the DNA in the sample. Duplication also assists in designation of a peak as an artefact. If there are several alleles in one or both reactions that are not duplicated then this may indicate gross sampling variation (including the possible presence of a mixture) or contamination of one of the duplicates

and the profile is not considered suitable for statistical interpretation due to this uncertainty.

In practice, many LCN profiles are mixtures. The use of a consensus necessarily limits the usefulness of the peak area data and, together with uncertainty about the designation of artefacts and potential drop-out or drop-in of alleles, restricts the statistical interpretation performed on many mixed profiles.

4 Intelligence Applications Involving STRs

The potential of STRs in the construction of intelligence databases was first realised with the SGM profiling system in the UK (Werrett, 1997). The national DNA database in England and Wales was launched April 10, 1995. Profiles obtained from individuals arrested and convicted of a recordable offence are entered on the database and compared against crime profiles arising from unsolved cases. The identification of a matching individual can provide the police with one or more potential suspects. In the UK, the FSS acts as custodian for the police, of the UK intelligence database; this now uses the more discriminating SGMPlus system. An analysis of the civil liberty and evidentiary issues for the UK can be found in Redmayne (1998). DNA Databases are now operational in many countries and efforts at harmonization of techniques have led to successful international exchanges of STR data; e.g. within Europe (Gill *et al.*, 2003).

The more extensive use of intelligence databases worldwide has prompted a debate on the most appropriate way to evaluate the DNA match evidence when the suspect has been identified from searching a database. The different approaches that have been advocated essentially correspond to differences in the propositions under consideration. For example, the NRC (1996), Morton (1997) and Stockmarr (1999) appear to be addressing the following alternatives:

- One of the individuals in the database left the crime stain.
- Some other person, unrelated to the defendant, and not in the database left the crime stain.

The corresponding LR is given by $1/Np$, where N denotes the number of profiles in the database and p denotes the match probability for an individual unrelated to the defendant.

Other authors (e.g. Balding & Donnelly, 1996; Donnelly & Friedman, 1999) consider a different pair of propositions:

- The defendant left the crime stain.
- Some other person, unrelated to the defendant, left the crime stain.

The LR in this case is slightly greater than $1/p$, which applies in the situation where a suspect has been identified by other means and then found to match the crime profile (sometimes called the *probable cause* scenario). This is because other possible suspects contained in the database have been eliminated as potential matches, in addition to the suspect being identified as a match.

Typically, evidence of a database match will be inadmissible in court and, hence, the FSS reports these cases using p, which will slightly understate the strength of the DNA evidence in favour of the defendant.

In cases where a suspect is excluded as a match to the crime profile, this can still provide valuable intelligence information if the mismatch is considered a “near miss”; i.e. the two profiles share a “large” number of alleles. This observation could be indicative of a blood relative of the suspect having left the crime stain. For example, statistical simulation studies (Foreman *et al.*, 2003) suggest that two full SGMPlus profiles sharing at least 12 or 13 alleles are much more likely if they originate from full siblings than unrelated people. Sjerps & Kloosterman (1999) describe a LR-based approach that evaluates the DNA profiling evidence given the following pair of propositions, say:

- The crime stain originates from a brother of the suspect.
- The crime stain originates from an unknown person, unrelated to the suspect.

The weight of the evidence is influenced by the number and rarity of the alleles shared by the profiles.

Ethnic inference is another area where DNA has provided a casework intelligence tool. Evett *et al.* (1992) addressed the problem using SLP evidence and their work has been extended to STR systems by Lowe *et al.* (2001). The underlying concept is reasonably intuitive and is based on estimates of the population proportion of the crime DNA profile within a range of different ethnic groups. For example, the DNA profiling evidence can inform the investigator in how to proceed with mass screens and enable evaluation given propositions of the sort:

- The DNA was left by an Afro-Caribbean male.
- The DNA was left by a SE Asian male.

This evaluation is based on the rarity of the STR profile and the circumstances of the case. The information is provided to the police for investigative purposes only.

5 A Framework for Interpreting DNA Evidence

5.1 Investigator – Evaluator

A useful model that helps scientists focus on their role is called the '*Investigator/Evaluator*' dichotomy. In reality, scientists operate in both *investigator* and *evaluator* modes in many of the cases that they examine. Providing opinion in these two different modes requires different mind-sets. An understanding of these differences is essential. Unless and until there is clarity about this distinction, there is a risk that misleading opinions could be given.

In *Investigator* mode, it is the scientist's role to form an inference from the observations. While attending a crime scene, or examining items associated with a victim, the police may put forward the following investigative questions:

- Is this a biological fluid and what is its exact nature?
- What potential set of actions may give rise to a stain of that nature in this quantity and at this location?

The scientist will form and communicate a 'best bet' to explain the observations based on his knowledge and experience. His opinion will be expressed as a probabilistic statement about the issue itself. Generally, scientists are operating in this mode before a suspect is arrested and charged with an offence. Nordby (2000) insists that the investigative stage integrates abduction, deduction and induction into a unified process but begins with abduction. Abductive reasoning follows a process of generating hypotheses, ranking hypotheses by their probability, testing hypotheses with new observations and eliminating or re-ranking the hypotheses. De Forest (1999) describes a process called 'hypothesis generation and cogitation'. The investigator arrives at a 'best bet' to explain the observations, refining that view as further observations are made. Opinions provide direction and options to the investigation. The issues addressed by the scientist tend to be centered on, or generated by, the crime.

We do not attempt to articulate fully the inferential steps of the investigator endeavour here but concentrate instead on the evaluator role that becomes predominant as the case progresses to court. In *Evaluator* mode, the scientist's role is to form a view on the weight of evidence to be assigned to scientific findings. This is the primary role of the scientist in what may be called *post-charge* cases; i.e. cases in which a suspect has been arrested and charged. In this role, there is a scientific consensus that the weight of evidence associated with DNA findings should be reported through a LR (Evett & Weir, 1998). This embodies the principle that interpretation of evidence is not possible unless one considers at least two competing propositions. In most cases in the adversarial system, these will represent respectively the prosecution and defence positions. This places emphasis on a

balanced and impartial view of the evidence. Although the practice for reporting DNA evidence is variable, there is a strong move towards the adoption of the likelihood ratio as a means of conveying the weight of DNA evidence (in Europe, for example, see Taroni *et al.* (2002)).

5.2 A Balanced Assessment of the Weight of the Evidence Using a Likelihood Ratio

Much has been written about the Bayesian view of evidence interpretation within the legal framework—see, for example, Robertson & Vignaux (1995) who provide an excellent overview followed by Redmayne (1996). A criminal trial can be viewed as two competing positions—prosecution and defence; Bayes' theorem then provides a logical structure for evaluating the evidence E which denotes the outcome of the DNA analysis:

$$\frac{\Pr\left(H_p|E,I\right)}{\Pr\left(H_d|E,I\right)} = \frac{\Pr\left(E|H_p,I\right)\Pr\left(H_p|I\right)}{\Pr\left(E|H_d,I\right)\Pr\left(H_d|I\right)}$$

where H_p denotes the proposition put forward by the prosecution, H_d denotes the proposition put forward by the defence and I denotes the framework of circumstances pertaining to the case in the form of time, location, witness statements describing the alleged actions, etc. It is important to emphasize that the framework will contain elements that are uncertain and liable to change. It represents the state of knowledge at a particular time in the criminal justice process. If the framework changes then the interpretation will have to be reviewed.

The forensic scientist seeks to assess the following LR:

$$LR = \frac{\Pr\left(E|H_p,I\right)}{\Pr\left(E|H_d,I\right)}.$$

Formulation of the LR crystallizes the key features of the interpretation of forensic science evidence. These have a crucial impact on the content of statements (Evett *et al.*, 2000c), namely: (1) the assessment of the LR is carried out within and is highly dependent upon a framework of circumstances; (2) the interpretation of evidence is essentially a relative measure that weighs the probability of the scientific observations assuming at least two competing propositions; and (3) The role of the scientist is to assign probabilities to the findings given each proposition considered. The latter feature prevents the scientist from transposing the conditional and committing the prosecutor's fallacy (Balding & Donnelly, 1994; Evett, 1995).

Hence, before any scientific finding can be evaluated and also, before any decision can be made on an examination strategy, it is imperative for the scientist to elicit the propositions in the case that stem from the relevant framework of circumstances in the case. This is the core concept of the *Case Assessment and Interpretation (CAI)* model (Cook *et al.*, 1998b). *CAI* fosters the view that interpretation and assessment must be at the forefront of the scientist's mind.

5.3 Hierarchy of Propositions

Managing the elicitation of propositions is without doubt the most difficult aspect of case interpretation. To assist the scientist in tackling this, it is convenient to visualize a 'hierarchy of propositions' as described in Cook *et al.* (1998a). In summary, the hierarchy has three levels: *offence, activity* and *source*. With the advent of more sensitive DNA techniques, it has become necessary to propose a further level, called 'sub-level 1', or 'sub-source level' (Evett *et al.*, 2000b). In DNA cases, we can illustrate the hierarchy through the following examples:

Case	*Case 1: Allegation of rape by a stranger*	*Case 2: Allegation of punching*
Framework of circumstances	Ms. V alleges she has been raped by a stranger on her way home one night. She described the man as a Caucasian male in his thirties, who speaks English with no accent. A male DNA profile was obtained from the semen recovered from the internal vaginal swab and loaded on the National DNA database. Mr. Smith matches the profile and becomes the main suspect. He denies any knowledge of the victim and the alleged incident.	Witnesses claim they saw Mr. V being punched by another white man. Mr. V sustained several cuts to his scalp. Mr. V states that he had been drinking with his attacker, a Mr. Smith, in a pub for six hours previous to the attack. Mr. Smith is arrested one hour after the incident. He agreed he was in Mr. V's company but denies the assault. The police have submitted Mr. Smith's jumper and reference blood samples from both men for laboratory examination.
Scientific findings	The male profile from the DNA mixture obtained from the semen recovered on the high vaginal swab from Ms. V matched the profile of Mr. Smith.	A presumed bloodstain on the sleeve of Mr. Smith's jumper gave a DNA profile matching Mr. V.
Offence level	H_p: Mr. Smith raped Ms. V. H_d: Some other man raped Ms. V.	H_p: Mr. Smith assaulted Mr. V. H_d: Some other man assaulted Mr.V.
Activity level	H_p: Mr. Smith had sexual intercourse with Ms. V. H_d: Some other man had sexual intercourse with Ms. V.	H_p: Mr. Smith punched Mr. V. H_d: Some other man punched Mr. V.
Source level	H_p: The semen stain came from Mr. Smith. H_d: The semen stain came from some other man.	H_p: The bloodstain came from Mr. V. H_d: The bloodstain came from some other man.
Sub-source level	H_p: The DNA profile obtained is a mixture originating from Mr. Smith and Ms. V. H_d: The DNA profile obtained is a mixture originating from an unknown male and Ms. V.	H_p: The DNA profile obtained came from Mr. V. H_d: The DNA profile obtained came from an unknown person.

Case	*Case 3: Stain (presumably blood) found at a burglary scene*	*Case 4: Cigarette end found at a burglary scene*
Framework of circumstances	A house has been burgled. Access was gained through a window that has been broken. Stains that appear to be blood have been recovered from the remaining glass fragments. It is thought the burglar bled at the scene from cuts obtained as a result of breaking the glass. On the same night, Mr. Smith became a suspect after presenting himself at the Hospital emergency department to be treated for cuts on his hand. Mr. Smith denies the burglary and ever being at the scene.	A house has been burgled. Near to the point of entry, on a paved patio, a scene of crime officer has recovered a cigarette end. None of the family living in the house are smokers and the patio area is swept regularly. A few weeks later, a suspect was apprehended in connection with another incident and his DNA matches the DNA found on the cigarette end. He denies ever being anywhere near the home that was burgled.
Scientific findings	The DNA profile obtained from the recovered stains matches Mr. Smith.	The DNA profile obtained from the cigarette end matched Mr. Smith.
Offence level	H_p: Mr. Smith is the burglar. H_d: Some other man is the burglar.	H_p: Mr. Smith is the burglar. H_d: Some other man is the burglar.
Activity level	H_p: Mr. Smith bled at the scene. H_d: Some other man bled at the scene.	H_p: Mr. Smith is the person who smoked the cigarette. H_d: Some other unknown person smoked the cigarette.
Source level	H_p: The bloodstain comes from Mr. Smith. H_d: The bloodstain comes from some other person.	H_p: The saliva on the cigarette end came from Mr. Smith. H_d: The saliva on the cigarette end came from some other man.
Sub-source level	H_p: The DNA profile is that of Mr. Smith. H_d: The DNA profile is that of an unknown person.	H_p: The DNA recovered from the cigarette end came from Mr. Smith. H_d: The DNA recovered from the cigarette end came from some unknown person.

It is not suggested that all case situations can be forced into this apparently rigid structure. However, it provides useful guidance when analyzing the issues within a case and identifying those areas in which a scientist can make a contribution. If we assume that full DNA profiles (at 10 loci, for example) are obtained in all four cases, then the LR for all pairs of propositions at sub-source level will be in the order of a billion, assuming that the unknown contributor is unrelated to the suspected contributor and that no laboratory error has occurred. The match probability for these pairs of propositions would simply be the inverse of the LR.

To progress from sub-source to source level, in addition to the match probability, we need to

consider uncertainties surrounding the relationship between the DNA profile obtained and the underlying body fluid assumed. This will depend on the type of presumptive tests conducted in an attempt to identify the body fluid, their specificity and sensitivity, the microscopic examination carried out, the efficiency of obtaining DNA profiles from targeted fluids, etc. Although in some cases the relationship between the DNA profile obtained and the presumed body fluid will be uncontroversial (for example, in Case 1 if sperm heads were observed), in others (such as Case 4), this progression may not be straightforward and will require expert knowledge.

To evaluate his findings within a framework of propositions at activity level, the scientist will need to take account of information on background DNA and issues of DNA transfer, persistence and recovery. This is in addition to the (sub-)source level information. An analytical approach to computing such LRs is given by Evett (1984). The LR will depend on the match probability but also on background and transfer probabilities and will be different from the one billion figure expressed at sub-source level. The scenario in Case 3 has been explored recently at activity level by Evett *et al.* (2002), resulting in a LR value of the order 1000.

The closer to offence level the expert is able to assess the evidence, the more fit-for-purpose the interpretation will be, providing greater added value to the court. However, it must be recognised that the amount of background information required from the framework of circumstances increases as the expert moves up the hierarchy from sub-source to activity level, regardless of his breadth of expertise.

In DNA cases, because of the ease with which match probabilities can be computed, there is a tendency for forensic scientists to assess their evidence at sub-source level. However, it is important for scientists to ask themselves how the case is to be progressed up the hierarchy. When this requires consideration of technical uncertainties that are clearly outside the general knowledge of a court or a jury (e.g. the probability of finding semen or blood given the alleged circumstances), the scientist has a duty to guide the court in its progression to offence level. Even if the scientist thinks that the uncertainties involved in this progression are such that no guidance can be given as to the assessment of the evidence in light of the alleged activities, then he still has a duty to inform the court in this regard; if not questioning the relevancy of the examination in the first place. For example, in Case 2 above, Mr. Smith may be claiming that he was standing close to Mr. V when the alleged assault took place and that this is when blood was transferred to his jumper. Hence, an examination solely to determine the blood DNA type would be completely irrelevant in helping address the activity-level propositions of whether or not it was Mr. Smith who punched Mr. V—a blood spatter analysis would probably be needed.

5.4 'No Comment' Interviews and a Staged Approach

When the framework of circumstances provides a sparsity of information or when the suspect gives a 'no comment' interview, the scientist may have difficulty in eliciting the propositions, especially those pertaining to the defence. In these cases, two options are open to the scientist:

(a) To report initially only factual information about the findings and highlight the need for a framework of circumstances to assess the evidence. When this framework is made available, then an evaluative statement can be produced.
(b) To construct propositions on behalf of the parties involved and assess the findings within this framework. For example, in Case 1, the scientist may assess the evidence under a defence proposition that the victim had sexual intercourse with another man, unknown and unrelated to Mr. Smith, although there may be nothing in the framework of circumstances made available to suggest that this unknown man should be unrelated.

5.5 Considering More Than Two Propositions

The above framework has been presented as if every evaluative case can be summarized by two mutually exclusive propositions representing, in the adversarial court system, the prosecution and defence views respectively. This is not always the case. For example, if we take case scenario 3 at sub-source level, the prosecution view will be along the lines of 'the DNA profile obtained originates from Mr. Smith' (H_0). On the other hand, the defence may allege the possibility that the profile originates from either an unknown unrelated person (H_1) or a brother of Mr. Smith (H_2). These cases cannot be handled easily using a LR approach because it is not possible to separate the likelihoods from the respective priors. It is necessary to implement the full form of Bayes' theorem to compute the posterior probability of H_0 (Balding, 2000).

$$\Pr(H_0|E, I) = \frac{\Pr(E|H_0, I)\Pr(H_0|I)}{\sum_i \Pr(E|H_i, I)\Pr(H_i|I)}.$$

We see in section 7 that such cases are efficiently handled using Bayesian Networks.

6 DNA Mixtures and Body Identification Using Mitochondrial DNA

In previous sections, the paper has focused on simple cases involving single crime stains matching a profiled suspect. Evidence interpretation in more complex situations is considered by Evett & Weir (1998). These include DNA mixtures originating from more than one person, paternity and other familial testing and cases involving the identification of remains of bodies. Probabilistic expert systems/Bayes' nets are emerging as very powerful tools for use in forensic interpretation. In particular, in facilitating the assessment of complex patterns of DNA evidence and family pedigrees such as arise in immigration cases and paternity/maternity testing. These ideas are discussed in the next section. We focus here on past and current practice followed in the interpretation of DNA mixtures and body identification (ID) cases.

6.1 DNA Mixtures

The main factor identifying a mixed sample is the presence of more than two bands or alleles at one or more of the loci. With STR profiles, the peak intensity information, represented by peak areas, is also used in terms of identifying excessive peak imbalance at a locus. In a heterozygote pair from one individual, the two alleles are expected to be of a similar height and area; so excessive differences are an indication of a mixed DNA sample.

One method of evaluating mixed profiles that is still adopted, primarily by forensic organisations in North America, calculates the probability of exclusion (PE) (Devlin, 1992). This represents the proportion of the population exhibiting a DNA band or allele that is not present in the mixed profile. The approach has severe limitations, one of which is that it takes no account of drop-out of contributor alleles; i.e. when alleles from a contributor fail to be detected. The National Research Council (1996) endorses the LR approach described in Evett *et al.* (1991) which utilises more of the available case-specific genetic information; e.g. the victim's or suspect's profile. The method involves consideration of the potential genotypes of unknown contributors given the prosecution and defence propositions.

The greater sensitivity of the SGMPlus and LCN techniques has led to an increase in cases involving mixtures; e.g. background DNA from unknown persons is often detected in practice. This has resulted in a range of papers in the literature addressing all aspects of the problem, many of which describe methodology that translates easily to computer code. For example, Weir *et al.* (1997) assume HWE and adopt a LR approach; they provide a computer program at
`http://www2.ncsu.edu/ncsu.CIL/stat_genetics/`. This is extended to take account of popu-

lation substructure within loci in Curran *et al.* (1999). The theory is extended further by Fung & Hu (2002) who address the case where contributors to the mixture originate from different ethnic groups. Fukshansky & Bär (2000) derive probability formulae that can be implemented when unknown contributors and profiled individuals are blood relatives.

All of the LR approaches discussed above are based solely on which alleles or peaks are present in the mixture. Therefore, if we consider the case of a 2-person mixture, where $\underline{a}$ denotes the vector of alleles observed at a locus, $\underline{G} = [(G_{11}, G_{12})(G_{21}, G_{22})]$ denotes the genotypes of the two contributors and H denotes the hypothesis then the LR is composed of terms of the type

$$p\left(\underline{a}|\underline{G}, H\right)\Pr\left(\underline{G}|H\right).$$

However, STR profiles exhibit additional information that is useful in interpreting a mixture, namely the peak intensities or areas. These are an indication of the amount of DNA being contributed by the allele to the mixture. Hence, the DNA evidence comprises two elements, $E = (\underline{a}, \underline{w})$, where $\underline{w}$ denotes the peak area vector corresponding to $\underline{a}$.

It is important to incorporate both elements of the evidence in any assessment of a mixture; peak areas provide valuable information as to the relative DNA contributions of the individuals present in the mixture. At the FSS, scientists currently take account of the peak area information, using it to assign weights of zero or one to possible genotype combinations for unknown contributors. The process is summarized in various publications (Clayton *et al.*, 1998; Gill *et al.*, 1998a,b; Gill, 2002).

Evett *et al.* (1998) developed a model for 2-person mixtures that takes account of the peak area evidence more formally by modelling ratios of peak areas using Normal probability distributions. This is achieved by considering functions of $\underline{w}$ for the full range of genotype configurations $\underline{G}$ that can occur at a locus; e.g. at a 3-peak locus, one possibility is that the two contributors exhibit a homozygote and a heterozygote. Data from unmixed profiles are used to estimate model parameters for the mixture-independent functions; these essentially model the process of *preferential amplification* which refers to the variability in peak area for the two peaks in a heterozygote. The mixture-dependent probability distributions depend on the mixing proportion m of the two contributors and their parameters are estimated using data from known, made-up mixtures.

A computer program was developed within the FSS to implement the above approach using SGMPlus data and extending the theory to take account of population substructure through Balding & Nichols recursive formula (Balding *et al.*, 1996). However, it is not obvious how the approach might be extended to consider more than two contributors. Perlin & Szabady (2001) have developed a deterministic approach to resolving the contributor profiles in a mixture for entry on and search against an intelligence database. This is achieved by a process of optimisation based on a least-squares-type metric following removal of the random PCR processes that act on STR profiles. Although providing an adequate solution for intelligence purposes, it is wholly inappropriate once the evaluative stage is reached.

6.2 Body ID Using Mitochondrial DNA

In recent years, STR profiling has helped in the identification of bodies in many cases. This essentially involves obtaining profiles from known blood relatives of the person suspected of being the body and constructing a family pedigree describing the relationships between the body and the profiled individuals given the propositions of interest. The pedigree and propositions will be specific to each case and facilitate the evaluation of a LR (Evett & Weir, 1998).

There are situations, however, when nuclear DNA techniques, such as STR profiling, fail. For example, when investigating small, degraded samples, originating from material such as bone and hair shafts. The preferred technique in these cases is often to obtain mitochondrial DNA (mtDNA) sequences from the samples that can then be compared (see Tully *et al.*, 2001 and references therein).

mtDNA differs from nuclear DNA in that it is inherited only through the maternal line and individuals who are maternally related, over an arbitrary number of generations, should share the same mtDNA. Although the primary use of mtDNA is still in cases of body ID, it is increasingly being applied in the form of ordinary trace evidence in criminal casework

mtDNA was one of the techniques used to identify the skeletons found in Ekaterinburg, Russia, in 1991, in the famous case that sought to establish whether or not they were the remains of the Romanov family. Some of the bones were confirmed as being from the last Tsar, Tsarina and three of their children by comparison with DNA from living maternal relatives, including the Duke of Edinburgh (Gill *et al.*, 1994).

In the forensic context, mtDNA sequences of chemical bases observed at a particular region are obtained. The statistical evaluation of such evidence involves the consideration of some issues that are different to those that impact on the interpretation of nuclear DNA. The main one is mutation. Mutation rates for bases occurring in mtDNA sequences are much greater than those observed for STR alleles and these can vary according to the forensic sample obtained; i.e. blood, hair etc. Hence, it is possible that samples obtained from maternally-related individuals or even the same person will produce different sequences. Tully *et al.* (2001) describe what options are available to forensic practitioners for calculating LRs; these depend crucially on the case circumstances and the nature and relevance of any data that are available to estimate required quantities; e.g. mutation rates.

7 The Use of Bayesian Networks to Interpret DNA Evidence

The use of graphical networks to represent legal issues is not new. Wigmore made extensive use of hierarchical representation to summarize the range of issues in legal cases (Wigmore, 1937), Friedman (1986a,b) introduced probabilistic networks and Kadane & Schum (1996) published a detailed graphical and probabilistic analysis of the Sacco and Vanzetti case. Early applications of probabilistic networks in forensic science were developed by Aitken and coworkers (Aitken & Gammerman, 1989; Aitken *et al.*, 1996) and Dawid & Evett (1997). Recent dedicated books (Jensen, 1996; Cowell *et al.*, 1999) and the availability of simple and intuitive computer packages to develop Bayesian networks, have promoted a steady growth of applications to forensic science problems. A list of software currently available can be found at
`htttp://www.ai.mit.edu/~murphyk/Software/bnsoft.html`.

A Bayesian network is a graphical model for expressing the probabilistic relationships among a set of variables. These graphical models combine probability theory and graph theory. In essence, a network consists of nodes, representing variables, and arrows that symbolize the conditional relationships between the variables. A variable might be a continuous variable, such as a measurement of a quantity, or a set of discrete outcomes, such as the truth or otherwise of a set of propositions. The model efficiently encodes the joint probability distribution for a large inter-related set of variables. It gives not only a graphical representation of the problem but enables probabilistic inference to be made. In the context of forensic science, Bayesian networks assist scientists not only in describing a complex problem but also in computing the effect of the scientific evidence on the plausibility of the relevant proposition. The simplest form of a Bayesian network dealing with DNA evidence applies when two source-level propositions are addressed:

H_p: The bloodstain has come from Mr. Smith.
H_d: The bloodstain has come from some unknown man, unrelated to Mr. Smith.

The corresponding Bayesian network is given below.

The two nodes (*Propositions* and *Evidence*) represent uncertain events. The directed arc, symbolized by the arrow, between the two nodes expresses a conditional relationship (causal in this case). In other words, the probabilities associated with the events of the *Evidence* node will be conditioned on the truth or otherwise of the events of the *Propositions* node. For this node (called a *child* node), probabilities are expressed in terms of conditional probability tables which depend on the *parent* node.

Such networks can be programmed very simply using software packages such as Hugin Lite (`http://www.hugin.com`). As soon as the evidence is observed (e.g. a full SGMPlus match), this is propagated back through the network to compute the posterior probability of H_p. The calculation relies on a simple application of Bayes' Theorem. If the prior odds are set to evens then the posterior odds (one billion) will be numerically equal to the LR that represents the weight of the scientific evidence. It is straightforward to extend the network to consider more than 2 propositions; for example:

H_0: The bloodstain has come from Mr. Smith.
H_1: The bloodstain has come from some unknown man, unrelated to Mr. Smith.
H_2: The bloodstain has come from a brother of Mr. Smith.

which results in the following:

The real power of Bayesian networks is in their ability to deal with increasing degrees of complexity, where a full algebraic solution is not feasible. For example, coping with the calculation of a LR, at sub-source level, when the result from the DNA analysis is a mixed profile from more than 2 persons is difficult (Weir *et al.*, 1997). Mortera and coworkers offer an elegant solution using Bayesian networks implemented in Hugin (Mortera, 2002; Mortera *et al.* to appear). As an alternative to Hugin, Green has developed a suite of functions in *R* called GRAPPA: `http://www.stats.bris.ac.uk/~peter/Grappa/`.

Networks that deal successfully with paternity disputes have also been developed by Dawid and co-workers using Hugin (Dawid *et al.*, 1999, 2002). Extending this research, Cowell has developed FINEX (Forensic Identification by Network EXpert systems), a general interactive computer program capable of analysing problems of disputed paternity. A given disputed paternity network may be constructed quickly using a graphical user interface with all input made using a mouse without the need for many local conditional probability tables (Cowell, submitted).

Addressing propositions at activity level force more detailed consideration of the nature of the relationship between the *Propositions* and the *Evidence* nodes. This will involve probabilities of *transfer* and *persistence*; also, the presence of *background* DNA material, unrelated to the alleged set of actions, must be taken account of. A generic Bayesian network can be drawn up as follows:

We have used Bayesian networks with this general structure to explore cases involving glass, fibres and saliva (DNA) evidence (Puch *et al.*, in preparation). Also, Evett *et al.* (2002) used Bayesian networks to interpret small quantities of DNA at activity level, with strong emphasis being placed on sensitivity analyses.

Moving up an extra step in the hierarchy—towards offence level—will require consideration of the relevance of the recovered evidence; i.e. the probabilistic association between the recovered DNA

material and the alleged offence. Garbolino & Taroni (2002) have explored scenarios, previously solved algebraically, involving bloodstains taking account of their relevance and fibre evidence. Aitken *et al.* (2002) have explored the evaluation of cross-transfer evidence in DNA cases.

As argued by Fenton & Neil (2000), with the BN approach it is possible to make all of the implications of Bayesian reasoning clear to lay people, without them having to understand any of the underlying mathematics or perform any of the calculations. The implications of this in the legal profession could be profound. In future Bayesian arguments could be presented to juries (and accepted by them) in just the same way as forensic evidence is now. Bayesian networks provide the mechanism to do this by offering a visual representation and graphical summary of the evidence.

8 The Future of DNA Interpretation

Since its debut as a forensic tool in the UK in 1986, DNA profiling methodology has advanced at a dramatic rate. Judicial reviews of old cases are frequently reported in the media. New techniques, in particular SGMPlus and LCN, offer the potential for obtaining DNA evidence from very small, old and degraded biological crime samples. It is therefore imperative that courts are provided with appropriate and balanced statistical interpretation of DNA results.

An increased awareness of the principles underlying the CAI model, discussed in Section 5, is essential in the forensic context. This applies not only to forensic scientists but to judges, barristers and police officers alike. Feedback from presentation of these ideas to the UK judiciary has been very positive; there was recognition of clear benefits to the criminal justice system in terms of impartiality, effectiveness and efficiency. From a practical perspective, there is a need for more extensive data collections in the future, in order to address activity-level propositions in the hierarchy. Furthermore, data relevant to the interpretation of LCN profiles is scarce. Experiments need to be designed to cover a range of situations met in practice and that will provide data to inform an assessment of issues such as contamination, transfer and persistence of DNA.

Interpretation of mixtures will remain important in the future, especially LCN mixtures and complex mixtures originating from more than two persons. Again, more comprehensive data collection will be required to facilitate the statistical modelling of these situations; specifically, the random processes that are involved in generating a profile from a DNA sample. Probabilistic expert systems and Bayesian networks will offer solutions to DNA mixtures and many more complex problems in the future. They represent one of the more significant advances in DNA interpretation in recent years since they have the ability to deal with highly complicated patterns of evidence through the independence of conditional probability distributions. User-friendly graphical representations facilitate discussion of complex DNA problems with non-statistical experts.

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Résumé

Cet article présente un inventaire des questions relatives à l'utilisation du profilage ADN dans la science légale. Une courte section historique décrit les principales étapes statistiques qui ont eu lieu pendant le rapide développement de la technologie ADN et ses utilisations opérationnelles. De plus grands détails sont ensuite donnés pour l'interprétation de questions sur les profils AND STR, ce qui inclut:

– les méthodes qui tiennent compte des sous-structures de population dans les calculs ADN;
– le travail conduit en parallèle par le Conseil de Recherche Nationale des Etats-Unis (NRC);
– l'évolution depuis les tests d'indépendance multiple vers des expériences qui démontrent la robustesse des procédures;
– la pratique contestable de l'attribution de source avec "certitude scientifique raisonnable";
– l'effet de l'interprétation des profils obtenus sous techniques de plus en plus sensibles, la technique LCN en particulier
– l'utilisation de profils ADN comme outil d'intelligence;
– l'interprétation de mélanges ADN.

L'expérience de ce type de preuve dans les tribunaux britanniques sera également présentée et commentée. L'article présentera un cavenas d'interprétation centré sur le concept de rapport de vraisemblance, inscrit dans une hiérarchie de propositions. Finalement, l'utilisation de réseaux Bayesien pour interpréter la preuve par ADN sera abordée.

Mots clefs: Criminalistique; Police scientifique; Interprétation de la preuve; Rapport de vraisemblance; Profile ADN; Mélange de profils ADN; Réseaux Bayesien.

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